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BooMondai: A Gamified Language Learning Platform Integrating User-Generated Quizzes and FSRS-Based Spaced Repetition for Language Learners in Davao City

PROPONENT 1, Loejee Miguel Dulaugon

PROPONENT 2, Daniel David Lupase

PROPONENT 3, Bernard Louie Pacanza

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General Terms: Terms

Additional Key Words and Phrases:

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1. INTRODUCTION

1.1 Background of the Study

The demand for language proficiency among Filipinos has grown substantially in recent years, driven by overseas employment programs, academic exchange initiatives, culture

industry growth and expanding trade relations with worldwide economies. From Japan alone, according to Tesda (n.d.), there are approximately 82 Total Courses, 23 Unique Institutions in 5 Davao Regions alone that have a TVI (Technical Vocational Institution) for Japanese. The Philippines Statistics Authority also did a survey that estimated the amount of overseas Filipino workers was estimated at 2.16 Million, previously it was 1.96 million the year before that (Mapa, 2024) Asia being 77.4% of OFWs (Mapa, 2024). For the past decade, access to quality instruction has remained concentrated in private institutions and urban centers.

For many learners without access to formal classes, self-study through digital tools represents the primary avenue for language acquisition. Learners that are able to have Learner Generated Digital Media (LDGM) all expressed satisfaction and a better attitude towards technology as students who are comfortable with making digital content are also more likely to want to use mobile learning apps (Bermudez et al., 2022).

In Davao City, this dynamic is particularly relevant. As one of the Philippines' largest and most economically active cities, Davao maintains established trade, cultural, and educational partnerships with other countries. The presence of foreign businesses, cultural organizations, and bilateral exchange programs has created a growing population of students and young professionals seeking practical language skills. However, formal language instruction in the city remains limited in availability and affordability. Many learners resort to free online tools or self-paced applications.

The current findings reveal that the effectiveness of gamified tools in Foreign Language Learning (FLL) was mixed. Some bring positive changes, while others show negative changes or no differences at all. (Luo, 2023). Research in mobile-assisted language learning (MALL) consistently identifies three factors as critical to successful language acquisition: motivation and engagement, opportunities for active recall, and mechanisms for long-term retention (Ishaq et

al., 2021; Shortt et al., 2021) . Gamified platforms such as Duolingo and Kahoot! address motivation effectively but rely on expert-curated content and do not incorporate scientifically validated retention scheduling, often times the spaced repetition methods focus their methods on predicting student memory or designing an optimal review schedule, thus contradicting the integrity of the spaced repetition system (Su et al., 2023). Spaced repetition systems such as Anki address long-term retention through the Free Spaced Repetition Scheduler (FSRS) algorithm as the approach reduces error by 64% (Su et al., 2023). However, it is solitary, non-gamified, and lacks social or community features. No existing platform meaningfully integrates all three components: gamified quizzing, user-generated content, and FSRS-based spaced repetition within a single accessible tool.

This study addresses that gap by designing and evaluating BooMondai, a platform where learners participate in user-generated quizzes and have their performance be automatically fed into a personal FSRS review deck. The platform is designed for foreign language learners, with the initial evaluation focused on learners in Davao City. It is aligned with SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities) by providing a free, accessible, and scientifically grounded language learning tool to a population underserved by existing solutions (United Nations, n.d.).

1.2 Problem Statement

Apps that use FSRS while implementing gamified features that address the lack of accessible, engaging and community-driven language-learning platforms are lacking in the market.

This Specific problems of the study are as follows:

1. Most existing mobile language learning platforms depend heavily on expert-curated content and structured progression paths, which limit adaptability and learner autonomy (Ishaq et al., 2021). Platforms such as Duolingo and Kahoot! do not allow learners to contribute quiz content or tailor materials to their specific vocabulary needs.
2. Spaced repetition tools such as Anki provide scientifically validated long-term retention through the FSRS algorithm but are solitary, non-gamified, and lack social or community features that sustain motivation and enjoyment over time (Zapata Vicente, 2023).
3. No empirical study has directly compared a gamified, community-driven FSRS-integrated platform against a conventional FSRS-only tool on measures of both vocabulary retention and learner experience specifically motivation, engagement, and enjoyment among foreign language learners in a Philippine context. Gamification has been shown to significantly influence intrinsic motivation and learning value among students, with learning value identified as the strongest predictor of students' intention to use a technology (Maragañas & Dioso, 2025; Ruedas & Pacatang, 2024)r .

1.3 Objectives

The main objective of this study is to design, develop, and evaluate BooMondai, a gamified language learning platform integrating user-generated quizzes and FSRS-based

spaced repetition, by comparing its effectiveness in vocabulary retention, motivation, engagement, and enjoyment against Anki among foreign language learners in Davao City.

The specific objectives are as follows:

1. To develop a web-based platform that supports To develop a web-based prototype of BooMondai that supports user-generated quiz creation with text, image, and audio enhancements, enabling single-player and optional multiplayer quiz participation with an optional pre-quiz vocabulary preview feature and automatic FSRS deck enrollment upon quiz completion.
2. To implement a two-system learning architecture in which quiz sessions serve as initial FSRS rating events using self-reported difficulty ratings (Again, Hard, Good, Easy) to seed each card's spaced repetition schedule and a separate FSRS review deck manages long-term retention through algorithmically scheduled follow-up sessions.
3. To evaluate the effect of BooMondai versus Anki on long-term vocabulary retention by comparing delayed retention test scores between Group A and Group B after two to four weeks.
4. To assess and compare learner motivation, engagement, and enjoyment between Group A and Group B at two time points immediately after the first session and at the end of the study period using validated Likert-scale instruments.
5. To assess learner perceptions of the vocabulary preview feature, FSRS review deck, platform usability, and user-generated content features among participants through post-study Likert-scale surveys and the System Usability Scale.

1.3.1 Evaluation Framework

In order to properly evaluate the effectiveness and expected outcomes of the study, we will be applying an evaluation framework.

Table I. Evaluation Framework Table.

Specific Aims	**Outcomes**	**Outcome Indicators**	**Data Source**	**When and by Whom**
Baseline — Day 1 (before study begins)				
Establish self-assessed language proficiency as a moderating variable	Baseline proficiency level	Mean Likert score and categorical proficiency level (None to Advanced); used to verify group equivalence after random assignment	Proficiency screener — Likert + categorical *(Group A: via prototype · Group B: Google Form)*	Day 1, before first session; self-administered by all participants
Establish learner interest in the target language as a covariate	Baseline language interest level	Mean Likert score ≥ 3.50 = moderate to high interest; used as covariate in retention analysis	Language interest — Likert *(Group A: via prototype · Group B: Google Form)*	Day 1, before first session; self-administered by all participants
Objective 1 — Short-term retention and experience				
Evaluate short-term vocabulary retention of BooMondai vs Anki	Immediate vocabulary recall after first session	Score on 30-item vocabulary test (Set A) — higher mean score in Group A vs Group B	Vocabulary test Set A *(Group A: via prototype · Group B: Google Form)*	Immediately after first session; self-administered by both groups

Specific Aims	Outcomes	Outcome Indicators	Data Source	When and by Whom
		indicates BooMondai advantage		
Assess and compare enjoyment, engagement, and motivation immediately after first session	Perceived enjoyment, engagement, and motivation at short-term	Mean Likert score ≥ 3.50 per dimension; compared between groups and against long-term administration	Enjoyment, engagement, motivation — Likert short-term *(Group A: via prototype · Group B: Google Form)*	Immediately after first session; self-administered by all participants
Objective 2 — Long-term retention and experience				
Evaluate long-term vocabulary retention of BooMondai vs Anki	Delayed vocabulary recall 2–4 weeks after first session	Score on 30-item vocabulary test (Set B) — significantly higher mean score in Group A vs Group B indicates BooMondai long-term advantage	Vocabulary test Set B *(Group A: via prototype · Group B: Google Form)*	1 week after final session; self-administered by both groups
Assess and compare enjoyment, engagement, and motivation after extended use	Perceived enjoyment, engagement, and motivation at long-term	Mean Likert score ≥ 3.50 per dimension; compared between groups and against short-term administration	Enjoyment, engagement, motivation — Likert long-term *(Group A: via prototype · Group B: Google Form)*	End of study period; self-administered by all participants
Objective 3 — Learner				

Specific Aims	Outcomes	Outcome Indicators	Data Source	When and by Whom
experience (end of study)				
Assess perceived usefulness of vocabulary preview *(Group A only)*	Perceived helpfulness of pre-quiz preview feature	Mean Likert score ≥ 3.50 = preview perceived as useful	Preview usefulness — Likert *(via prototype, Group A only)*	End of study; self-administered by Group A only
Assess perceived usefulness of FSRS review deck *(Group A only)*	Perceived helpfulness of spaced repetition review	Mean Likert score ≥ 3.50 = FSRS perceived as useful	FSRS usefulness — Likert *(via prototype, Group A only)*	End of study; self-administered by Group A only
Assess perceived value of user-generated content features	Learner interest in creating personal decks and using community content based on actual experience	Mean Likert score ≥ 3.50 = positive perception of UGC features	UGC perception — Likert *(Group A: via prototype · Group B: Google Form)*	End of study; self-administered by all participants
Assess overall platform usability *(Group A only)*	Usability and satisfaction with BooMondai prototype	SUS composite score ≥ 68 = above average usability; findings inform pre-release improvements	System Usability Scale (SUS) *(via prototype, Group A only)*	End of study; self-administered by Group A only

1.4 Significance of the Study

This study provides value to multiple stakeholder groups while contributing to the empirical literature on gamified spaced repetition in mobile-assisted language learning.

For language learners in Davao City. BooMondai offers a free, accessible alternative to formal language instruction for language learners. By combining community-generated content with scientifically validated retention scheduling, the platform addresses both the engagement gap and the forgetting problem that characterize existing self-study tools. Learners at any proficiency level can contribute to and benefit from a shared pool of quiz materials without financial barriers.

For language learners in Davao City. BooMondai offers a free and accessible alternative to both formal language instruction and existing self-study tools. By combining gamified quiz mechanics, vocabulary preview, and FSRS-based spaced repetition, the platform addresses both the engagement limitations of Anki and the retention limitations of gamified platforms such as Kahoot! and Duolingo. Learners of Japanese, Chinese, and other foreign languages gain a tool that is scientifically grounded, socially engaging, and locally accessible.

For educators and language instructors. The platform can supplement formal instruction by providing a structured, gamified review tool that students can use independently. The FSRS review deck in particular provides a self-directed retention mechanism that extends the learning beyond classroom hours. Instructors may also use the platform's quiz creation feature to design course-specific vocabulary sets for their students.

For educational technology researchers. This study contributes empirical evidence on whether a gamified FSRS-integrated platform produces comparable or superior retention outcomes to a conventional FSRS tool, while also measuring the motivational and experiential

dimensions of each approach. The direct comparison between BooMondai and Anki fills a gap in the MALL literature that has not been addressed in prior research.

For the academic field and SDG alignment. The study contributes to SDG 4 (Quality Education) by developing an accessible, evidence-based learning tool for an underserved learner population, and to SDG 10 (Reduced Inequalities) by reducing barriers to language education in a resource-limited context. As a student-developed capstone project, it documents a locally grounded, research-oriented response to an identified educational need in Davao City.

For future developers. The system architecture, FSRS integration approach, and evaluation findings documented in this study may serve as a reference for developers building similar gamified spaced repetition platforms for other languages, subjects, or learner populations.

1.5 Scope and Limitations

Scope of the Study

This study covers the design, development, and evaluation of BooMondai, a web-based language learning platform for Japanese and Chinese learners. The platform includes the following features:

- User-generated quiz creation with text, image, and audio support, accessible to registered users.
- Optional pre-quiz vocabulary preview, allowing users to review all question-answer pairs before beginning the quiz session to establish a basis of familiarity with the target vocabulary.
- In-quiz difficulty rating (Again, Hard, Good, Easy) prompted upon each correct answer, which seeds the initial FSRS scheduling parameter for that card.
- Automatic enrollment of all quiz items into the user's personal FSRS review deck upon quiz batch completion.
- A separate FSRS review deck for scheduled long-term retention practice, independent of the quiz session system.
- Leaderboard system tracking retention streaks and review consistency, with optional data visibility controls allowing users to hide specific performance metrics.
- Quiz batching in sets of 10 to 20 questions for large quiz sets.

The study includes planning, design, development, and evaluation phases. A quasi-experimental evaluation will compare vocabulary retention between a quiz-plus-FSRS group and a quiz-only control group, involving 30 respondents from Davao City over two to four weeks.

Limitations of the Study

Language Problems Limitation. While the platform is designed to support any language pair, the evaluation of this study is limited to Japanese and Chinese language learners in Davao City, as these represent the most in-demand foreign languages for employment in the

region, Asia being 77.4% of OFWs (Mapa, 2024). Findings may not generalize to learners of other languages or to learners in other geographic contexts.

Language skills coverage. The quiz-based format focuses on reading and listening comprehension through text, image, and audio materials. The application does not assess speaking or writing production skills, which require different types of learning activities.

Language proficiency measurement. This study does not aim to measure long-term improvements in language proficiency. Determining whether the application produces measurable learning gains would require longitudinal research beyond the capstone project timeline.

Content quality and moderation. The platform relies on user-generated content, which may vary in quality and accuracy. The initial version does not include a formal moderation system for reviewing quiz content before publication. This limitation may affect the reliability of learning materials available on the platform.

Technical constraints. The application will be hosted on a standard cloud platform with limited server resources. Performance may be affected during simultaneous usage by multiple users. The application is optimized for modern web browsers on desktop and mobile devices, but compatibility with older browsers or devices with limited specifications is not guaranteed. Basic security measures such as input validation and user authentication are implemented, but the platform does not include advanced security features suitable for large-scale deployment.

Real-time implementation. Due to the time constraints and limited resources. The real-time multiplayer feature is not plausible and will not come with the prototype.

Evaluation scope. Usability testing will involve 20 respondents from Davao City who are currently learning or have interest in learning Japanese or Chinese. Testing will be conducted over a period of two weeks, capturing initial user reactions and feedback after using the application. This sample size and duration may not represent the broader population of Filipino language learners or capture long-term usage patterns.

Feature limitations for future development. Planned features such as optional monetization for content creators, advanced progress tracking, and expanded language pairs are outside the scope of this initial version and reserved for future development work.

Generalizability of findings. Because the study focuses specifically on Japanese and Chinese learners in Davao City, the findings may not directly apply to other languages or learner populations in different regions.

Full Long-Term Time Constraint. Due to the study only being available to conduct for 4 months, the full effect of the long-term retention (1 year) is not possible.

2. REVIEW OF RELATED LITERATURE

Related Literature

The proliferation of mobile-assisted language learning (MALL) has revolutionized how individuals acquire new languages, particularly through the incorporation of gamification, spaced repetition systems (SRS), user-generated content (UGC), and real-time interactive elements. This review synthesizes key empirical studies, systematic reviews, and platform analyses on these components, emphasizing their application to vocabulary building, comprehension skills,

and learner motivation—especially relevant for character-based languages such as Japanese and Chinese. It draws from research on prominent platforms like Duolingo (gamified lessons), Anki (SRS flashcards), community-driven tools, and Kahoot! (real-time quizzes), while pinpointing gaps that the proposed capstone project—a community-driven app enabling user-generated, real-time multiplayer quizzes—aims to fill.

2.1 The Free In FSRS: The Active Development Of The Community

FSRS stands for Free Spaced Repetition Algorithm. It is an open-source project backed by machine learning techniques. Its open source characteristic allows it to be continuously improved throughout the years without the need of profit margins while taking improvements directly from the community. Open source software (OSS) has played a fundamental role in technology and our social lives. Because of the various advantages of OSS, more and more companies are participating extensively in open source development (Li et al., 2025). According to Hoffman, et al. (2024), the demand side value of OSS is estimated to be $8.8 trillion. Firms would need to spend 3.5 times more on software if it were not free. This emphasizes the value of open source as OSS today also includes cutting edge technologies in various fields.

2.2 FSRS The Science Based Algorithm: Research Backed Long Term Memory Project

An algorithm that is optimized for the dynamics of memory has been a popular topic by language learners for memorizing and learning words and sentence structure. Most spaced repetition systems focus on predicting student recall or by creating an optimal review schedule (Su et al., 2023). FSRS proposes a powerful machine learning algorithm for memorization. This algorithm is optimized for long term memory “combines psychologically proven theories (e.g., forgetting curve and spacing effect) with modern

machine learning techniques to reduce the cost of learners in forming long-term memory" (Su et al., 2023). Which does not address short term memory. Anki, which is the main app FSRS was developed for, is reported to lack engaging content (Zapata Vicente, 2023). The spaced repetition approach only aims for long-term retention, which is more feasible for learning something that takes a long amount of time than for situations where trying to learn information that is needed on demand or "short-term". Watching the video repeatedly at different intervals with spaced repetition creates boredom with the learners which leads them to abandon the learning environment (Schimanke et al., 2023)

2.3 Gamification: The Motivation Component

Multiple researches from the Davao Region shows that gamification is a way to make learning more engaging and motivating as it makes the students feel that learning is enjoyable (Ruedas & Pacatang, 2024). A month's use of gamification shows a marked improvement in academic and engagement levels. Engagement scores improved indicating a satisfactory level of student involvement which improves retention and contributes to student satisfaction which creates a more meaningful and enjoyable experience (Maragañas & Dioso, 2025). This capstone's quiz question types are also based on Duolingo, while excluding Duolingo's spaced repetition algorithm "Half Life Regression" (HLR) which is less effective than Anki's FSRS (Su et al., 2023). It has a significant positive impact on the proficiency of their English language skills.

(Bermudez et al., 2022; Calolo et al., 2025, 2025; Dreisbach & Demeterio, 2021; Lim & Arcilla Jr, 2021; Naparan & Bacasmot, 2024; Tagaro & Jr, 2025)

2.4 User Generated Content (UGC): The Personalized Learning Experience

A study conducted in Panabo City, Davao Del Norte about Learner Generated Digital Media (LDGM) all rated themselves high when it came to “Attitude towards technology”, “Understanding Assignments”, “Satisfaction” and “Performance Expectancy” as students who are comfortable making digital content are also more likely to want to use mobile learning apps (Bermudez et al., 2022). Many e-learning environments also have a problem of lacking content, which makes UGC a promising approach as there is a huge amount of content freely available on the internet. However one of the most important issues when it comes to UGC is the lack of quality assurance, a solution that for this challenge can be derived from the way Wikipedia works which is through the management of content by the community itself (Schimanke et al., 2023).

2.5 Local First: UGC Is Owned By The User and Addressing Internet Connection Problems

Another problem remains with Mobile Assisted Language Learning (MALL) is that apps are usually online, this creates an issue specially in the Davao Region in Philippines where due to the lack of a complete infrastructure of the internet, leads to the slow loading of content or inaccessibility of it (Calolo et al., 2025; Lim & Arcilla Jr, 2021; Maragañas & Dioso, 2025; Naparan & Bacasmot, 2024; Tagaro & Jr, 2025). Local First technology solves these issues. By storing the data in an accessible format located in every device that has the software installed, it prioritizes offline use first which solves the problem of having to use the internet every time the user accesses information. It also has additional benefits like If a service shuts down, the data goes along with it, this approach allows the users to keep the data on their devices while maintaining privacy

and long-term preservation, and this study also shows that it can be beneficial for asynchronous learning (Otsamo, 2025).

3. Methodology

3.1 Purpose Of Methodology

This research was conducted in order to determine whether a gamified learning app, with user generated quizzes and FSRS would play a significant role in long-term and short-term retention for memory as well as if the gamification elements would be engaging, motivating and enjoyable for the participants. In order to complete these research goals, The researcher conducted this research with 20 respondents. Participants used an app prototype that gathers their data to a database.

3.2 Research Design

This research is a quantitative quasi-experimental study that measures using pre-test and post-test scores, and the use of the likert scale in order to assess if the participants feel engaged and motivated using the app.

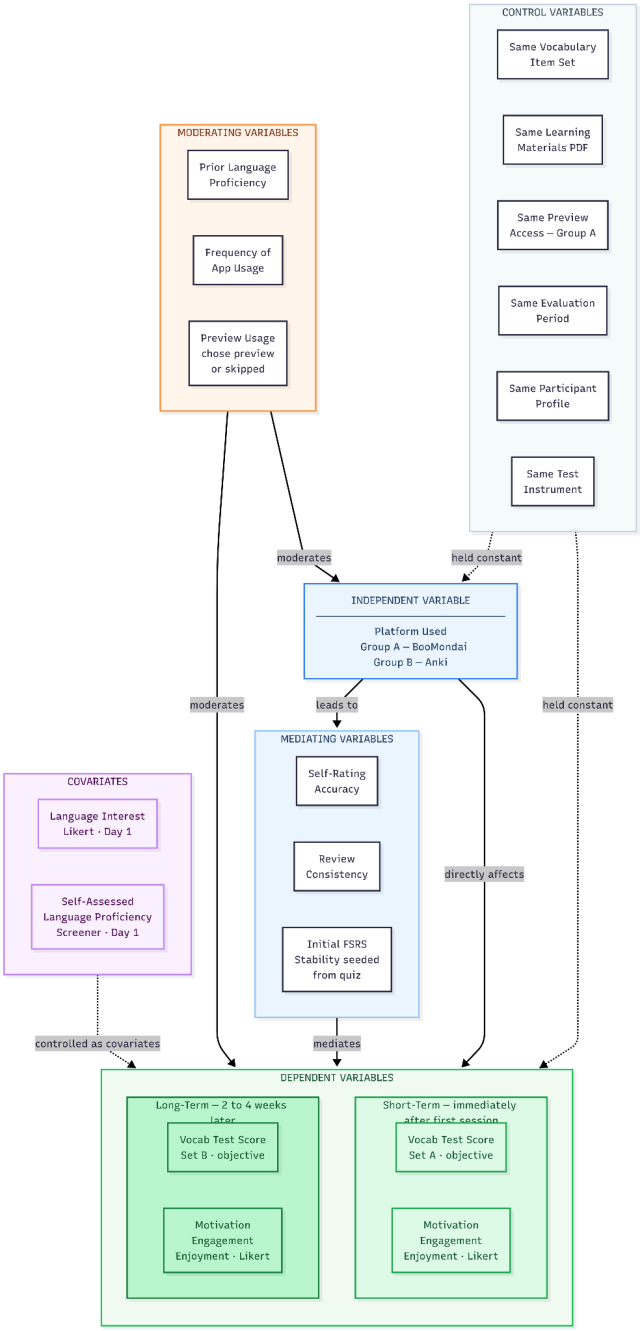

Fig. 1. Conceptual Framework Flowchart.

The conceptual framework of this study illustrates the relationships between the key variables involved in comparing BooMondai against Anki as a language learning tool for foreign language learners in Davao City. The framework identifies one independent variable, two sets of

dependent variables, two covariates, and three categories of supporting variables that contextualize the primary comparison.

The independent variable is the platform used, which is the sole condition that distinguishes the two groups. Group A uses BooMondai, a gamified web-based platform featuring optional vocabulary preview, a quiz session with FSRS self-rating mechanics, leaderboards, streaks, and user-generated content. Group B uses Anki, a conventional FSRS-based flashcard tool that provides spaced repetition scheduling without gamification, social features, or quiz mechanics. Both groups study the same researcher-provided vocabulary set delivered as a premade deck on their respective platforms.

The dependent variables are organized into two time points to capture both immediate and sustained effects of the platform. Short-term dependent variables, measured immediately after the first session, include the vocabulary test score from Set A as an objective measure of immediate recall, and Likert-scale ratings of motivation, engagement, and enjoyment. Long-term dependent variables, measured two to four weeks later, include the vocabulary test score from Set B as a measure of delayed retention, and the same three Likert dimensions administered a second time to capture whether learner experience changes over extended platform use. Measuring both objective scores and subjective experience at two points allows the study to determine not only whether BooMondai improves retention but also whether it sustains motivation and enjoyment better than Anki over time.

Two covariates are measured on Day 1 before the study begins. Language interest captures each learner's pre-existing motivation toward the target language, while the self-assessed proficiency screener establishes their baseline knowledge level. Both are statistically controlled in the retention analysis because learners with stronger interest or higher

prior proficiency may retain vocabulary more effectively regardless of which platform they use. Controlling for these factors strengthens the internal validity of the group comparison.

Three categories of additional variables complete the framework. Moderating variables prior language proficiency, frequency of app usage, and preview usage may strengthen or weaken the relationship between the platform used and the observed outcomes, and are monitored across the study period. Mediating variables self-rating accuracy, review consistency, and initial FSRS stability seeded from quiz performance represent the mechanisms through which BooMondai is expected to produce its effects on retention and learner experience. Control variables including the same vocabulary item set, premade deck, evaluation period, participant profile, and test instrument are held constant across both groups to ensure that any observed differences in outcomes can be attributed to the platform rather than external factors.

3.3 Data Gathering

Before the researchers began gathering data they sought permission from the head of the institution. At the start of the data gathering, participants are prompted the Data Privacy Act of 2012 to give consent to the use of their data This study employed a purposive sampling method. The participants are picked with the following qualifications; interested in our language set (Japanese, Chinese), based in davao city, has internet access, willing to participate for 2-4 weeks. The primary means of data collection was through an application prototype that sends data to an online database, only accessible by the researchers and participants. This study involves the participation of 20 participants. The researchers adapted the following survey questionnaires to measure the level of the variables: Gamification and Learning Engagement

Questionnaire (GLEQ) by Cahyani (2016) similarly to Casia et al. (2024). Data was gathered through online and offline means. Throughout the data gathering process, participants were regularly checked in order to maintain the quality and validity of the data.

3.4 Data Analysis

The data gathered from the vocabulary retention tests, Likert-scale surveys, proficiency screener, and System Usability Scale will be analyzed using a combination of descriptive and inferential statistical methods. All statistical analyses will be performed using SPSS or an equivalent statistical software package.

Descriptive Statistics. Descriptive statistics will be computed for all instruments administered throughout the study. Mean scores and standard deviations will be calculated for each Likert-scale dimension motivation, engagement, enjoyment, language interest, perceived usefulness of preview, perceived usefulness of FSRS, and perceived value of UGC features —for both groups and at each time point. Percentage scores will be reported for individual vocabulary test items to identify which items produced the strongest and weakest recall across both groups. SUS scores will be computed using the standard SUS scoring formula and interpreted against the established benchmark of 68, above which a system is considered to have above-average usability.

Independent Samples t-Test. An independent samples t-test will be used to determine whether statistically significant differences exist between Group A (BooMondai) and Group B (Anki) on the following measures: short-term vocabulary retention scores from Set A, long-term vocabulary retention scores from Set B, and mean Likert scores for motivation, engagement, and enjoyment at both short-term and long-term administrations. The null hypothesis for each

comparison states that there is no significant difference between the two groups. A significance level of $p < 0.05$ will be used as the threshold for rejecting the null hypothesis. If the assumption of normality is not met due to sample size constraints, the Mann-Whitney U test will be applied as a non-parametric alternative for the affected comparisons.

Paired Samples t-Test. A paired samples t-test will be conducted within each group to compare short-term and long-term Likert scores for motivation, engagement, and enjoyment. This analysis will determine whether learner experience significantly changed over the course of the study period for Group A and Group B independently. A significance level of $p < 0.05$ will be applied.

Analysis of Covariance (ANCOVA). To account for pre-existing differences in language proficiency and language interest between participants, an Analysis of Covariance will be conducted on the vocabulary retention scores. Language interest, measured through the Day 1 Likert survey, and self-assessed language proficiency, measured through the Day 1 proficiency screener, will be entered as covariates. The ANCOVA will determine whether significant differences in retention outcomes between groups persist after statistically controlling for these covariates, thereby strengthening the internal validity of the retention comparison.

Interpretation of Results. Likert-scale scores will be interpreted using the following scale: a mean of 1.00 to 1.79 is Very Low, 1.80 to 2.59 is Low, 2.60 to 3.39 is Moderate, 3.40 to 4.19 is High, and 4.20 to 5.00 is Very High. A mean score of 3.50 or above is used as the threshold for acceptable learner experience across all Likert dimensions. Vocabulary retention scores will be reported as raw scores out of 30 and as percentages, with gain scores computed where applicable to measure improvement between administrations.

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